UTKARSH RAJ

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EDUCATION

Master of Computer Applications (MCA) *Currently Pursuing*
Noida Institute of Engineering & Technology
Greater Noida, Uttar Pradesh

Bachelor of Science – Information Technology (B.Sc. IT) *July 2022 – July 2025*
Anugrah Narayan College (Patliputra University)
Patna, Bihar

PROJECTS

Sentiment Analysis of Movie Reviews using ML

Python | Scikit-learn | TF-IDF | MySQL | Pandas | Matplotlib

- **Developed** a binary **sentiment classifier** on **50,000 IMDB movie reviews** using NLP preprocessing (HTML tag removal, stopword filtering, **TF-IDF vectorization**) to prepare text data for ML modeling.
- **Trained and evaluated** 3 ML models (Logistic Regression, Naive Bayes, Linear SVM); achieved best **accuracy of 90.6%** and **F1-Score of 90.7%** with SVM on an 80/20 train-test split.
- **Deployed** final model (89.9% accuracy) using **Pickle serialization**, enabling real-time sentiment prediction on unseen reviews.

Airbnb Data Analysis

Python | Pandas | NumPy | Matplotlib | Seaborn | Plotly | EDA

- **Cleaned and transformed** Airbnb listings dataset — handled missing values, removed duplicates, parsed currency fields — ensuring **data integrity** for accurate analysis.
- **Designed** 13 business-driven visualizations (histograms, box plots, scatter plots, heatmaps) to analyze **pricing trends, room type distribution**, and neighbourhood-wise **revenue potential**.
- **Identified** top 10 high-revenue neighbourhoods and top 10 hosts, delivering **actionable pricing and marketing recommendations** for Airbnb hosts.

E-Commerce Sales Data Analysis (Superstore)

Python | Pandas | NumPy | Matplotlib | Seaborn | EDA

- **Performed** 22+ end-to-end **EDA analyses** on Superstore dataset across 3 product categories covering **monthly sales trends, profit margins**, and regional performance.
- **Applied** Pareto Analysis (80/20 Rule) to identify top revenue-driving products and measured **average shipping time by mode** — enabling supply chain and inventory optimization insights.
- **Uncovered** negative **discount-profit correlation** via pair plots and identified **top 10 loss-making products**, enabling targeted recommendations to reduce revenue leakage.

TECHNICAL SKILLS

Languages: Python, C++, SQL
Machine Learning: Scikit-learn, TF-IDF Vectorization, Model Training & Evaluation, Pickle Deployment
Data Analysis & EDA: Pandas, NumPy, Matplotlib, Seaborn, Plotly
Visualization & BI: Power BI, MS Excel (Pivot Tables, Charts)
Databases: MySQL
Tools & Version Control: Git, GitHub, Jupyter Notebook, VS Code
Soft Skills: Analytical Thinking, Problem Solving, Communication, Teamwork, Self-Learning

CERTIFICATIONS & ACHIEVEMENTS

- **Completed** Python, Data Science, Machine Learning, and SQL courses on **Infosys Springboard**.
- **Solved 100+ SQL problems on LeetCode**, strengthening **database querying and optimization** skills.

LANGUAGES

Hindi (Native) | **English** (Professional Proficiency)